



# AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

## TEST REPORT NO. 2013 - 60



**KATO Centrifugal Pump coupled to a KATO  
KAT186 Diesel Engine  
Model : KC40**

### KATO KC40 Centrifugal Pump coupled to a KATO KAT186 Diesel Engine

|                        |   |                                                                              |
|------------------------|---|------------------------------------------------------------------------------|
| Test Report No.        | : | 2013 - 60                                                                    |
| Date of Test           | : | June 22, 2013                                                                |
| Test Valid Until       | : | June 22, 2018                                                                |
| Brand                  | : | KATO                                                                         |
| Model                  | : | KC40                                                                         |
| Serial No.             | : | 2010530                                                                      |
| Type and Size          | : | Single-end suction, 100 x 100 mm,<br>Non-Self Priming Centrifugal pump       |
| Manufacturer           | : | KATO METAL CORP.<br>Valenzuela City                                          |
| Test Requested by      | : | TRAMAT MERCANTILE, INC.<br>747-749 Sabino Padilla, St., Sta. Cruz,<br>Manila |
| Machine Classification | : | Commercial                                                                   |

### SUMMARY

The KATO KC40 centrifugal pump coupled to a KATO KAT186 diesel engine was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The engine and pump both used double groove B-section pulleys with actual diameters of 762 mm (3.0 inches) and 102 mm (4.0 inches), respectively and two pieces of V-belts. The engine and pump were mounted on a steel base.

The Maximum Total Head (H) was 29.6 m at zero discharge condition. The shaft speed of the pump and engine at this point were 2581 and 3286 rpm, respectively with Maximum Engine Fuel Consumption Rate of 5.0 L/h. The Maximum Discharge Capacity of 30.0 L/s occurred at 8.6 m Total Head with Fuel Consumption Rate of 3.6 L/h. The shaft speed of the pump and engine at this point were 1689 and 2152 rpm respectively. The Maximum System Efficiency of 7.8% occurred at 25.2 L/s Discharge Capacity.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 8.0 m. The Net Positive Suction Head Available at this point was 1.8 m.

### PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pump performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. The pump was mounted on the laboratory test set-up shown in **Figure 1**.

The test was conducted on June 22, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna.



Figure 1. KATO KC40 Centrifugal pump coupled to KATO KAT186 Diesel engine on the test set-up

## METHOD OF TEST

The pumpset was tested following the “Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test”.

## DESCRIPTION OF THE PUMP

The KATO KC40 Centrifugal pump coupled to KATO KAT186 Diesel Engine shown in **Figure 2**, was a single-end suction, 100 mm x 100 mm, centrifugal pump coupled to a 7.5 kW (10 Hp) KATO KAT186 diesel engine. The pump and engine were mounted on a steel base. The power from the engine was transmitted to the pump shaft through a double groove B-section pulley and V-belts. The actual diameter of the engine and pump pulleys were 762 mm (3.0 inches) and 102 mm (4.0 inches) respectively. The suction inlet and discharge outlet were threaded and designed for a 100 mm diameter pipe. The impeller was a closed type made of brass material connected to the pump shaft through a thread. It was housed in a cast iron volute casing. The priming inlet was located on top of the pump beside the discharge outlet. The discharge outlet was located on top of the pump facing upward. The stuffing box used a packing seal. The pump shaft rotated on ball bearing kept in place by a bearing housing integral to the casing.

The prime mover, a 7.5 kW KATO KAT186 diesel engine was a four-stroke, air-cooled, single vertical cylinder diesel engine. Fuel feed system was of gravity type with the fuel tank located on top of the engine. It utilized a dual type air cleaner with pleated paper element and foam element moistened with oil. Lubrication system was force-feed lubricating type. Cooling was done by air forced by a fan/flywheel encased in a metal shroud. It was started by a rope-recoil system. A muffler was used in the exhaust system. **Table 1** shows the detailed specifications of the pump.



Figure 2. KATO KC40 Centrifugal Pump coupled to a KATO KAT186 Diesel Engine



### PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the KATO KC40 Centrifugal Pump coupled to KATO KAT186 Diesel Engine

|      |                                                                    |   |                                                       |
|------|--------------------------------------------------------------------|---|-------------------------------------------------------|
| 1.   | Brand                                                              | : | KATO                                                  |
| 2.   | Model                                                              | : | KC40                                                  |
| 3.   | Serial No.                                                         | : | 2010530                                               |
| 4.   | Make                                                               | : | KATO METAL CORP.<br>Valenzuela City                   |
| 5.   | Type                                                               | : | Single-end suction, Non-self priming Centrifugal pump |
| 6.   | Size (mm)                                                          | : | 100 x 100                                             |
| 7.   | Overall Dimension and Weight (Including mounting frame and engine) |   |                                                       |
| 7.1  | Length (mm)                                                        | : | 1215                                                  |
| 7.2  | Width (mm)                                                         | : | 445                                                   |
| 7.3  | Height (mm)                                                        | : | 600                                                   |
| 7.4  | Weight (kg)                                                        | : | 98                                                    |
| 8.   | Impeller                                                           |   |                                                       |
| 8.1  | Type                                                               | : | Closed                                                |
| 8.2  | Diameter (mm)                                                      | : | 175                                                   |
| 8.3  | Width (mm)                                                         | : | 37                                                    |
| 8.4  | No. of vanes                                                       | : | 8                                                     |
| 8.5  | Material                                                           | : | Brass                                                 |
| 9.   | Suction Inlet                                                      |   |                                                       |
| 9.1  | Type                                                               | : | Threaded flange                                       |
| 9.2  | Diameter (mm)                                                      | : | 100                                                   |
| 9.3  | Material                                                           | : | Cast Iron                                             |
| 10.  | Discharge Outlet                                                   |   |                                                       |
| 10.1 | Type                                                               | : | Threaded flange                                       |
| 10.2 | Diameter (mm)                                                      | : | 100                                                   |
| 10.3 | Material                                                           | : | Cast Iron                                             |

Table 1. (Cont'n)

|       |                                        |   |                                                            |
|-------|----------------------------------------|---|------------------------------------------------------------|
| 11.   | Casing                                 |   |                                                            |
| 11.1  | Type                                   | : | Volute                                                     |
| 11.2  | Material                               | : | Cast Iron                                                  |
| 12.   | Stuffing box                           | : | Packing Seal                                               |
| 13.   | Rotation                               | : | Clockwise when facing the suction inlet                    |
| 14.   | Rated speed (rpm)                      | : | 2400                                                       |
| 15.   | Rated Maximum Discharge Capacity (L/s) | : | 27.8                                                       |
| 16.   | Rated Maximum Total Head (m)           | : | 18.3                                                       |
| 17.   | Primemover (based on Plate Rating)     |   |                                                            |
| 17.1  | Brand                                  | : | KATO                                                       |
| 17.2  | Model                                  | : | KAT186                                                     |
| 17.3  | Serial No.                             | : | HP12100864                                                 |
| 17.4  | Make                                   | : | No Data                                                    |
| 17.5  | Type                                   | : | Single cylinder, 4-stroke, direct injection, diesel engine |
| 17.6  | Rated Output Power (kW)                |   |                                                            |
|       | Maximum                                | : | 7.5 @ 3600 rpm                                             |
|       | Continuous                             | : | No Data                                                    |
| 17.8  | Bore x Stroke (mm)                     | : | No Data                                                    |
| 17.9  | Displacement Volume (cc)               | : | 406                                                        |
| 17.10 | Cooling System                         | : | Air-cooled                                                 |
| 17.11 | Lubricating System                     | : | Force-feed                                                 |
| 17.12 | Starting System                        | : | Rope-recoil                                                |
| 17.13 | Air Cleaner                            | : | Dual type (Foam and pleated paper element)                 |
| 17.14 | Exhaust System                         | : | Muffler                                                    |

## PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pumpset Tested : KATO KC40 coupled to KATO KAT186 diesel engine

Date of Test : June 22, 2013

Test Condition

Ambient Temperature (°C) : 30.1  
 Atmospheric Pressure (mb) : 1013  
 Pumping Water Temperature (°C) : 31.8

Table 2. Results of Performance Test

| ENGINE SHAFT SPEED (rpm) | PUMP SHAFT SPEED (rpm) | FUEL CONSUMPTION (L/h) | TOTAL HEAD (m) | DISCHARGE CAPACITY (L/s) | WATER POWER (kW) | INPUT POWER (kW) | SYSTEM EFFICIENCY (%) |
|--------------------------|------------------------|------------------------|----------------|--------------------------|------------------|------------------|-----------------------|
| 3286                     | 2581                   | 5.0                    | 29.6           | 0.0                      | 0.0              | 52.1             | 0.0                   |
| 3108                     | 2434                   | 4.9                    | 26.3           | 5.4                      | 1.4              | 50.5             | 2.8                   |
| 2939                     | 2303                   | 4.6                    | 23.7           | 8.7                      | 2.0              | 47.8             | 4.2                   |
| 2788                     | 2164                   | 4.5                    | 20.9           | 11.9                     | 2.4              | 46.5             | 5.2                   |
| 2626                     | 2045                   | 4.1                    | 18.2           | 15.0                     | 2.7              | 43.1             | 6.2                   |
| 2478                     | 1930                   | 4.1                    | 15.7           | 18.4                     | 2.8              | 42.2             | 6.7                   |
| 2327                     | 1817                   | 3.8                    | 13.2           | 21.7                     | 2.8              | 39.6             | 7.1                   |
| 2221                     | 1734                   | 3.3                    | 10.7           | 25.2                     | 2.7              | 33.8             | 7.8                   |
| 2152                     | 1689                   | 3.6                    | 8.6            | 30.0                     | 2.5              | 37.1             | 6.8                   |



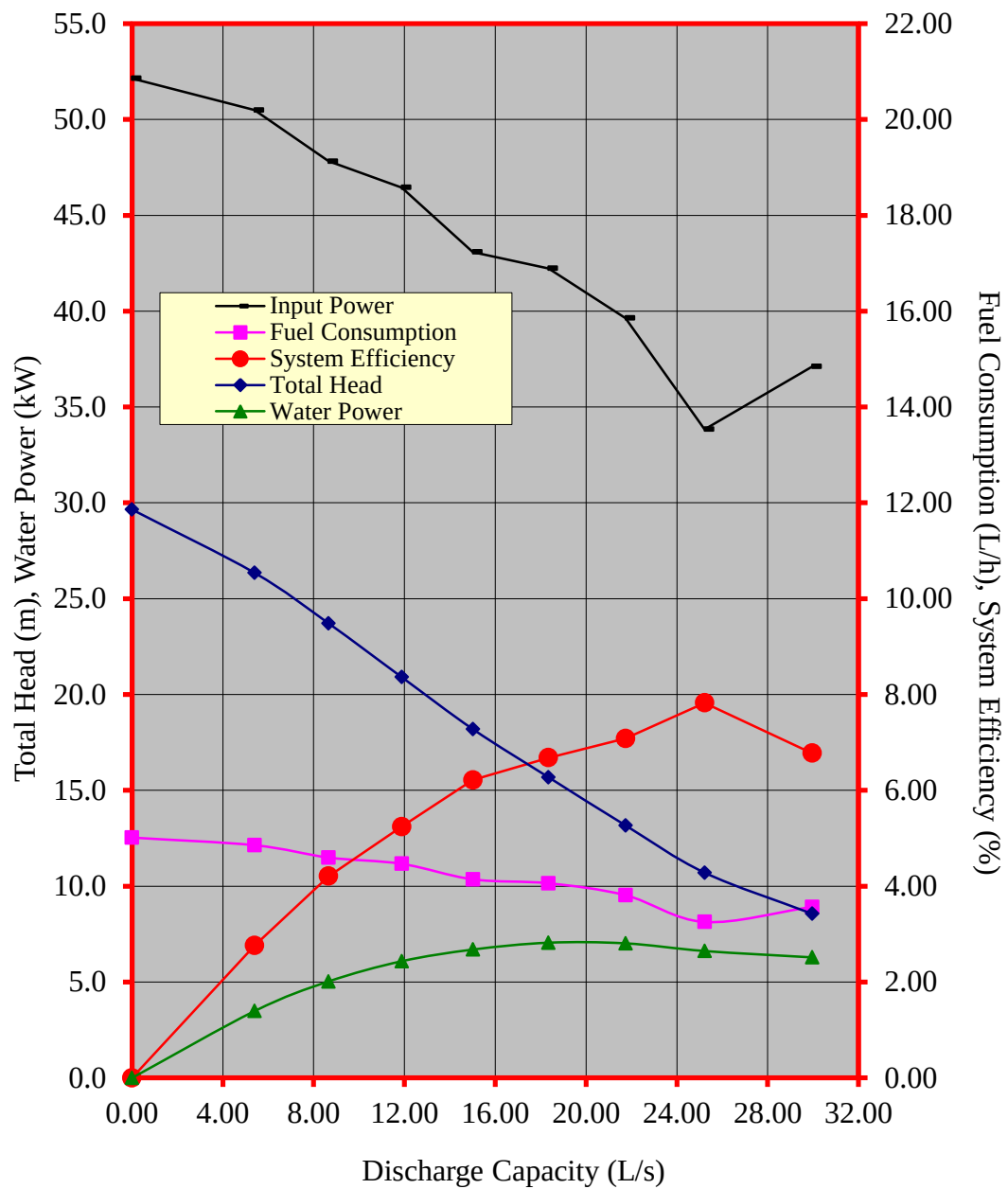


Figure 3. Performance characteristic curves of the KATO KC40 coupled to KATO KAT186 diesel engine centrifugal pump

### CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pumpset Tested : KATO KC40T coupled to KATO KAT186 diesel engine

Date of Test : June 22, 2013

Test Condition

Ambient Temperature (°C) : 31.1

Atmospheric Pressure (mb) : 1013

Pumping Water Temperature (°C) : 32.8

Table 3. Results of Cavitation Test

| ENGINE<br>SHAFT<br>SPEED<br>(rpm) | PUMP<br>SHAFT<br>SPEED<br>(rpm) | TOTAL<br>SUCTION<br>LIFT<br>(m) | NET<br>POSITIVE<br>SUCTION<br>HEAD<br>AVAILABLE<br>(m) | TOTAL<br>HEAD<br>(m) | DISCHARGE<br>CAPACITY<br>(L/s) |
|-----------------------------------|---------------------------------|---------------------------------|--------------------------------------------------------|----------------------|--------------------------------|
| 2072                              | 1609                            | 0.8                             | 9.0                                                    | 8.4                  | 30.2                           |
| 2152                              | 1690                            | 1.2                             | 8.6                                                    | 8.8                  | 29.7                           |
| 2185                              | 1698                            | 2.7                             | 7.2                                                    | 9.9                  | 27.6                           |
| 2215                              | 1705                            | 4.1                             | 5.7                                                    | 10.9                 | 25.3                           |
| 2276                              | 1728                            | 5.6                             | 4.3                                                    | 12.0                 | 23.2                           |
| 2688                              | 2107                            | 7.2                             | 2.6                                                    | 12.3                 | 10.8                           |
| 3137                              | 2463                            | 8.0                             | 1.8                                                    | 12.4                 | 0                              |

## DATA COMPUTATION

### ANNEX 1. Formula used during calculation and testing

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#### 1. Total Head (m), H

$$TDH = H_d - H_s$$

Where:  $H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

$H_d$  = Total Discharge Head (m)

$H_s$  = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$  = Pressure gage reading on the discharge pipe at  
gage connection converted to meter

$\frac{P_1}{\gamma}$  = Vacuum gage reading on the suction pipe at the  
point of gage connection converted to meter

$\frac{V_2^2}{2g}$  = Velocity head on the discharge pipe converted  
to meter. Computed by determining the velocity  
of liquid on the pipe by formula,  $V = Q/A$  where  
 $V$ , velocity of liquid inside the pipe in meter per  
per second,  $Q$ , Discharge rate of the pump in  
liters per second, and  $A$ , Cross-sectional area  
of the pipe in square meter.

$\frac{V_1^2}{2g}$  = Velocity head on the suction pipe converted to  
meter

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## Annex (Cont'n)

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$Z_2$  = The distance from the center of the pressure gage face to the centerline of the pump in meters

$Z_1$  = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity,  $Q$ , in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.
3. Water Power,  $WP$ , is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power,  $IP$ , is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determined by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

$T$  = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

$N$  = Input shaft angular speed in RPM

5. Pump Efficiency,  $\eta$ ,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available,  $NPSHA$ . This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

$P_{atm}$  = Atmospheric pressure during the test converted to meters

$P_{vp}$  = Vapor pressure of the test liquid at certain temperature converted to meters

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N.B.

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